SoilSidekick Pro® / LeafEngines™ SDK

Technical Overview for AFRL/RYMS Evaluation

March 2026 | Distribution: AFRL Evaluation Only

Mission

The LeafEngines™ SDK delivers micro-local environmental intelligence through a unified, privacy-preserving API stack. It enables partners to integrate real-time soil, vegetation, and environmental sensing into existing systems without ML expertise, and supports fully offline operational modes for contested, denied, or bandwidth-limited environments.

Core Capabilities

Multi-Modal Environmental Intelligence

- Soil composition, moisture, compaction, and structure inference
- Vegetation health with species-level indicators and stress detection
- Micro-climate and terrain-aware contextual reasoning
- Sensor-agnostic ingestion (RGB, NIR, multispectral, geospatial context)

Unified API Surface

- Single endpoint for all inference tasks with standardized outputs
- Regulatory-grade reasoning trace (transparent, auditable outputs)
- Supports batch, streaming, and edge-device workflows

Developer-Friendly Integration

- SDKs for iOS, Android, Web, and Python — integrates in hours, not weeks
- Lightweight, serverless-optional architecture; no specialized hardware required
- SOC 2–aligned security with role-based access control

Architecture Overview

LeafEngines™ acts as the enterprise-grade API layer embedding SoilSidekick Pro® capabilities into a pipeline:

1. Input: Image(s), sensor metadata, geolocation, optional auxiliary data
2. Pre-processing: On-device or hybrid (privacy-preserving, no raw data leaves device)
3. Inference: Patent-protected multi-modal reasoning engine
4. Output: Structured environmental intelligence with confidence intervals and uncertainty bounds
5. Sync: Offline caching with delayed sync or air-gapped deployment

Operational Modes

Offline-First	Full functionality without network connectivity. All inference executes on-device.
Hybrid	Local inference + cloud enhancement when available. Graceful degradation on disconnect.
Edge / Air-Gap	Low-power, low-bandwidth environments. Optional on-prem or air-gapped deployment.

Security & Compliance

- Privacy-preserving by design — no raw imagery required to leave the device
- SOC 2–aligned controls with comprehensive audit logging
- Patent-protected inference pipeline (no model weights or algorithms exposed)
- Optional on-prem or air-gapped deployment pathways

DV005 Objective Alignment

Mapping SDK Capabilities to AFRL/RYMS DV005 Requirements

DV005 Objective	SDK Capability	IP Protection	TRL	Status
1. Geo-Specific Scene Construction	Vincenty geodesy engine computes sub-meter displacements on WGS-84 ellipsoid. AlphaEarth 64-dim semantic embeddings at 10m resolution provide ground-truth environmental priors.	CIP #19/544,827 Claim 1, S106	6	□
2. Sensor-Independent Multimodal Output	EO/IR and RF modalities rendered as co-registered projections of a single environmental state vector (moisture, NDVI, conductivity). Shared physical priors enforce consistency.	Parent #19/320,727 + CIP S102	5–6	□
3. Cross-Modal Physical Consistency	Complementary filter (α=0.96) blends high-frequency inertial orientation with low-frequency magnetometer corrections. Real-time Pearson correlation validates cross-modal coherence.	CIP Claim 1, S102/S112	6	□
4. Uncertainty Propagation	Kalman-inspired variance model propagates positional uncertainty per step. Reliability gating prevents low-confidence data from persisting to storage or downstream consumers.	CIP Claim 1, S108/S110/S112	6	□
5. DIL-Resilient Offline Execution	100% deterministic math executes on-device with zero network dependencies. Exportable as standalone HTML. Three-tier fallback: GPS → Sensor DR → Seeded Centroid.	CIP Claim 1, S114 (Gating)	6	□
6. Distributed RF Sensing Support	Geodesic triangulation via Vincenty engine supports multi-node RF geolocation. Per-node uncertainty envelopes enable weighted fusion for distributed imaging arrays.	Parent #19/320,727 (Grid) + CIP S106/S108	4–5	⚠

Response: RYMS-Relevant Sensing Modalities

Per AFRL SME request, the following modalities and outputs are directly applicable to RYMS priorities:

Modality	SDK Implementation	RYMS Application
EO/IR Co-Registration	AlphaEarth 64-dimensional semantic embeddings encode land meaning (cover, moisture, vegetation health) at $10m^2$ resolution from multi-temporal Sentinel-2 imagery.	Synthetic scene generation with physics-grounded ground truth
RF Geolocation	Vincenty geodesic engine provides sub-meter triangulation for multi-node positioning. Per-node uncertainty envelopes support weighted fusion.	Distributed RF imaging array coordination
Uncertainty-Gated Outputs	Kalman-inspired variance propagation with write-inhibition at 500m uncertainty radius. All outputs carry confidence intervals.	Reliability-aware synthetic training data for ML pipelines

Proposed Next Steps

6. AFRL SMEs provide mission alignment feedback on prioritized modalities

7. SoilSidekick Pro delivers interactive offline demo (standalone HTML) for independent RYMS evaluation
8. Technical onboarding session: API walkthrough, sample integration, and field test design
9. Define minimum viable experiment (MVE): 30-min GPS-denied field test comparing DR-estimated vs. ground-truth endpoints

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